LoRa and WiFi at 2.4 GHz: a Cross-Technology Interference Evaluation

Ruana Saduakhas, Yerassyl Kadirzhanov, Dimitrios Zorbas
Nazarbayev University, School of Engineering & Digital Sciences, Astana, Kazakhstan
Authors' email: {firstname.lastname}@nu.edu.kz

***Abstract*—LoRa technology operating in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band presents an appealing solution for industrial and smart city deployments due to its global availability and compatibility. However, this frequency overlap raises concerns about cross-technology interference (CTI), particularly between LoRa transmissions and ubiquitous WiFi networks. This paper presents a comprehensive empirical study evaluating the interference effects between LoRa and WiFi under real-world conditions. Using off-the-shelf devices, we assess the impact of two WiFi modulations on LoRa packet delivery ratio across different LoRa settings. Our results reveal that while LoRa exhibits robustness against WiFi interference, certain configurations, especially high data rate modes, are vulnerable. Moreover, the results indicate that a LoRa packet can be successfully received if the WiFi received signal power is not higher than 25 dB than the LoRa one, on average. Finally, devoting more bits for error correction improves packet reception up to 23% in the conducted experiments.**

***Index Terms*—Internet of Things, LoRa, WiFi, Cross-Technology Interference**

I. Introduction

The rapid expansion of wireless communication technologies has led to increasing congestion in the unlicensed 2.4 GHz ISM band, where various protocols such as WiFi (IEEE 802.11), Bluetooth, Zigbee, and LoRa operate. The LoRa technology, initially designed for sub-GHz frequency bands, has recently been extended to the 2.4 GHz band, raising concerns about potential CTI when operating alongside other technologies occupying the same spectrum [1]. This paper particularly focuses on characterizing the interference and coexistence of WiFi and LoRa in the 2.4 GHz spectrum employing field experiments.

WiFi is a high-throughput technology using Orthogonal Frequency-Division Multiplexing (OFDM) modulation and Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) for medium access, operating on wide channels ($\geq$20 MHz). In contrast, LoRa uses Chirp Spread Spectrum (CSS) for long-range, low-power communication, with narrowband signals and pure ALOHA access. Although Semtech's application note [2] suggests LoRa has significant immunity to WiFi interference in controlled settings, the extent of this resilience remains unclear in real-world deployments. Given the fundamental differences in how these technologies operate in the 2.4 GHz band, it is essential to study their coexistence and mutual impact on performance.

The 2.4 GHz ISM band is heavily used and often crowded with both non-RF interference sources and various wireless technologies [3]. Deploying networks in this band requires careful frequency planning and consideration of nearby emitters [4]. While many technologies operating here are robust—thanks to high bandwidth, short airtime, and efficient Medium Access Control (MAC) strategies—LoRa, with its long airtime and narrowband channels, may be more susceptible to interference. To assess this, we conducted an experimental study using off-the-shelf devices. We measured LoRa performance at five specific distances from the receiver, analyzing how interference characteristics, such as proximity and modulation type, affect packet reception ratio (PRR). Additional experimental parameters are detailed later in the paper.

This study focuses on characterizing the impact of background WiFi traffic on LoRa communications. The reverse scenario of how LoRa affects WiFi is not considered, as WiFi networks typically have higher redundancy through multiple access points and more robust error recovery via the TCP/IP stack. In contrast, LoRa often depends on a single link and lacks built-in mechanisms for reliable retransmission, making it more vulnerable to interference.

The contribution of this work is that it presents a systematic experimental evaluation of WiFi interference on LoRa communication in the 2.4 GHz band, analyzing how modulation type, interference strength, and distance affect packet reception. It introduces a measurement-based methodology to quantify the resilience of LoRa under cross-technology interference and correlates performance with Signal-to-Interference-plus-Noise Ratio (SINR) conditions. As an additional contribution, the study provides a publicly available dataset containing RSS and Signal-to-Noise-Ratio (SNR) measurements across various distances, offering a valuable resource for future modeling and simulation efforts.

The remainder of this paper is organized as follows: Section II reviews related work on CTI and LoRa interference. Section III outlines the fundamental communication characteristics of both LoRa and WiFi. Section IV details the experimental setup and methodology used to capture CTI effects. Section V presents the results and analyzes their implications on network performance. Finally, Section VI summarizes the findings and outlines directions for future research.

II. Related Work

In sub-GHz bands, LoRa is typically deployed in unlicensed spectrum, which is also shared with other low-power wide-area network (LPWAN) technologies, industrial devices, and proprietary wireless systems. Unlike the 2.4 GHz band, where spectrum is heavily congested, sub-GHz bands tend to be less crowded but more fragmented, with varying regulations and duty cycle constraints depending on the region. Studies such as [5] and [6] have analytically shown that packet collisions due to overlapping transmissions are a key limiting factor in LoRa scalability, especially in dense deployments. Furthermore, coexistence challenges arise not only from intra-technology interference (e.g., overlapping LoRa networks) [7], but also from inter-technology interference with protocols such as Sigfox and Z-Wave [8, 9], IEEE 802.15.4g [10], and even certain types of wireless metering and alarm systems [11].

Besides, inter-technology interference in the 2.4 GHz ISM band has been widely studied due to the coexistence of multiple wireless technologies, such as WiFi, Bluetooth, and IEEE 802.15.4-based technologies (Zigbee, 6TiSCH etc.). These technologies often operate concurrently in close proximity and share overlapping spectrum channels, leading to significant performance degradation due to mutual interference [12, 3]. Several studies have examined the nature and impact of such interference. For instance, Bluetooth and WiFi, despite employing frequency hopping and spread spectrum techniques respectively, can suffer packet loss and increased latency in crowded environments [13, 14]. IEEE 802.15.4 networks have been shown to experience throughput reductions and unstable links in the presence of WiFi transmissions, especially when low-power IoT devices are involved [12, 15].

More recently, researchers have explored how LoRa at 2.4 GHz coexists with legacy technologies. Polak et al. [16] have shown that LoRa shows strong resistance to Wi-Fi interference, especially under co-channel conditions. Moreover, LoRa is more affected by the Complementary Code Keying (CCK) modulation of WiFi—employed in IEEE 802.11b—than by the OFDM-based one, employed in IEEE 802.11n. The robustness increases with higher SFs and lower bandwidths. However, the results are based on a spectrum analyzer output, not on real field experiments. Moreover, not all SFs were tested, while the channel bandwidths refer to the sub-GHz variant. While tested against Bluetooth [17], the authors found out a similar behavior; LoRa signals show higher immunity when SFs are higher and channel bandwidths lower. Finally, another research shows very high immunity of LoRa against LTE when both operate in the 2.4 GHz band [18].

III. Background

In this section, we briefly describe the communication characteristics of LoRa and WiFi in the 2.4 GHz spectrum and underline the principles behind in-band interference between the two technologies.

A. Spectrum overlap

LoRa and IEEE 802.11 both operate within the 2.4 GHz ISM band, sharing a significant part of the radio spectrum with other technologies such as Bluetooth and IEEE 802.15.4. As described in the LoRa physical layer specification[1], a LoRa network allows flexible channel configurations. However, three default channels must be implemented to support network Join Request messages. As illustrated in Fig. 1, there is a significant overlap between commonly used WiFi channels and LoRa default channels [19]. The complete overlap of the wide bandwidth of WiFi with the narrowband LoRa increases the likelihood of CTI. This makes LoRa more vulnerable to interference from signals within the same spectrum. This interference can cause packet collisions, and as a consequence, loss of information and higher energy consumption. Therefore, understanding the CTI levels is important for time-critical and resource-limited applications.

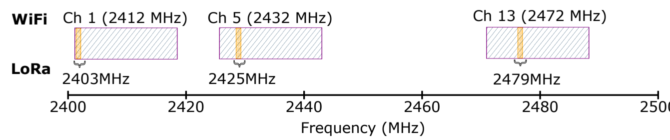


Fig. 1. Spectrum overlap between channels 1, 5, and 13 of IEEE 802.11 (hashed rectangles), and typical LoRa 2.4 GHz channels (solid yellow areas).

B. Physical Layer Characteristics

At the physical layer, LoRa employs a proprietary modulation technique developed by Semtech based on the known CSS modulation. CSS modulation uses chirp signals, which are frequency sweeps transmitted over the duration of a symbol. An increasing frequency sweep is called an "up-chirp", while the decreasing frequency is "down-chirp". This approach provides significant advantages that allow LoRa signals to be successfully received and demodulated even below the noise floor (a signal with negative Signal-to-Noise-Ratio (SNR)). In contrast, WiFi primarily uses OFDM that divides the frequency spectrum into subcarriers for efficient spectrum utilization and higher throughput. Due to the fundamental differences in modulation techniques, direct cross-technology communication between LoRa and WiFi is practically infeasible, unless a chirp and a LoRa frame emulation mechanism is provided [20]. Nonetheless, each modulation technique independently provides robustness to certain adverse channel conditions.

C. Data Layer Characteristics

At the MAC layer, LoRa-based protocols usually transmit packets without explicit scheduling or carrier-sensing mechanisms. The only exception is the Channel Activity Detection (CAD) feature, which identifies preambles of signals with identical SFs [21]. WiFi, in contrast, uses CSMA, often supplemented with Request-to-Send/Clear-to-Send (RTS/CTS) mechanisms to mitigate collisions. Consequently, one could expect that intensive WiFi traffic would significantly impact LoRa communications primarily due to density of WiFi transmissions, often overlapping LoRa transmissions. Since, LoRa lacks channel activity detection against WiFi traffic, the packet reception ratio at LoRa receivers is expected to decrease substantially.

D. Interference Mathematical Model

In this subsection, we mathematically model CTI between LoRa and WiFi assuming a log-distance path-loss signal

[1]https://www.semtech.com/uploads/technology/LoRa/phy-layer-2g4.pdf

attenuation model [22]. By estimating SINR thresholds for successful reception, this model helps determine minimum separation distances between transmitters and receivers, guiding the spatial layout and device positioning to ensure meaningful and controlled experimental conditions.

The received power of a LoRa signal at a distance d from the transmitter is given by:

$$P_r^{\text{LoRa}} = P_t^{\text{LoRa}} + G_t^{\text{LoRa}} + G_r^{\text{LoRa}} - PL_{\text{LoRa}}, \quad (1)$$

where P_t^{LoRa} is the transmission power of LoRa, G_t^{LoRa} is the LoRa antenna transmission gain, G_r^{LoRa} is the LoRa antenna reception gain, and $PL^{\text{LoRa}}(d)$ is the log-distance path loss of the LoRa transmission, given as follows:

$$PL^{\text{LoRa}}(d) = PL^{\text{LoRa}}(d_0) + 10\gamma^{\text{LoRa}} \log_{10}\left(\frac{d}{d_0}\right), \quad (2)$$

where $PL^{\text{LoRa}}(d_0)$ is the reference path loss at distance d_0, and γ^{LoRa} is the path-loss exponent.

Similarly, the received power of a WiFi signal at the LoRa receiver is:

$$P_r^{\text{WiFi}}(d') = P_t^{\text{WiFi}} + G_t^{\text{WiFi}} + G_r^{\text{LoRa}} - PL^{\text{WiFi}}(d'), \quad (3)$$

where P_t^{WiFi}, G_t^{WiFi}, and $PL^{\text{WiFi}}(d')$ are the corresponding WiFi transmission power, transmission antenna gain, and log-distance path loss. d' is the distance between the WiFi transmitter and the LoRa receiver, and d'_0 is the reference distance of the WiFi model.

The SINR at the LoRa receiver is calculated as follows:

$$SINR = \frac{P_r^{\text{LoRa}}}{I_{\text{WiFi}} + N}, \quad (4)$$

where $I_{\text{WiFi}} = P_r^{\text{WiFi}}$ is the WiFi power at the LoRa receiver, and $N = -174 + 10\log_{10} B$ is the thermal noise power, which depends on channel bandwidth B. The higher the SINR value, the stronger the signal, and thus, the higher the probability of receiving a packet.

Eq. (4) indicates that the longer the distance between a WiFi transmitter and a LoRa receiver, the lower the interference at the latter one. This interfering distance depends on the environmental conditions which affect the received power as well as on the immunity of LoRa against interference. SINR is estimated in Section IV-B.

IV. Methodology & Experiments Setup

This section outlines the experimental approach utilized to assess CTI between LoRa and WiFi. Detailed description of the equipment and employed methodology is provided.

A. Devices Description

To conduct experiments, we employed commercially available ESP32-based Adafruit Feather V2 devices with integrated IEEE 802.11 transceivers. Additionally, devices were equipped with SX1280 LoRa transceivers interfaced via Serial Peripheral Interface (SPI), as illustrated in Fig. 2. We used the same microcontroller units (MCU) and antenna to maintain hardware uniformity across all transmitting and receiving devices. WiFi devices used the same MCU and antenna, without the LoRa module.

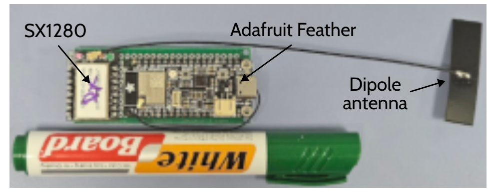


Fig. 2. Device used in the experiments: ESP32 equipped with an SX1280 module, and dipole antenna.

B. Preliminary experiments

To better understand the link budgets of both LoRa and WiFi in an indoor environment—and thereby drive the selection of suitable experiment locations—a series of preliminary measurements were conducted. These initial experiments followed a well-established procedure from the literature [23] to empirically determine the path loss exponent γ for both the LoRa and WiFi configurations. Estimating γ in the specific deployment environment allows for more accurate modeling of signal attenuation, enabling better planning for coverage and reliable interference analysis. The list of parameters used and obtained from these experiments are listed in Table I.

TABLE I
Obtained path-loss model parameters.

Parameter	**WiFi**	**LoRa**
Transmission power	P_t^{WiFi}=20dBm	P_t^{LoRa}=13dBm
TX/RX antenna gains	G_t^{WiFi}=2.5dBi	$G_t^{\text{LoRa}} = G_r^{\text{LoRa}}$=5dBi
Reference distance d_0	10m	10m
Path loss at reference distance	PL^{WiFi}=101dB	PL^{LoRa}=75dB
Thermal noise bandwidth		B=812kHz
Thermal noise power		N=-114.90dBm
Obtained path loss exponent	γ^{WiFi}=3.44	γ^{LoRa}=2.36

Fig. 3 illustrates SINR for variable distances using path-loss values obtained using settings of the preliminary experiments and the theoretical model. Taking into account an at least 20 dB CTI threshold [16] and a minimum SINR threshold of 6 dB to receive a LoRa signal [24], an estimated transition distance—above which all packets can be received—is around 3.4 meters. The results also indicate that there is a high collision rate below half meter, indicated with red, while distances between 0.5m and 3.4m may be affected depending on the SF and the LoRa immunity.

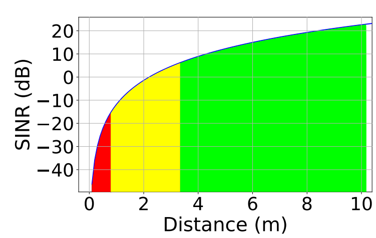


Fig. 3. SINR over variable distances between a LoRa receiver and a WiFi transmitter.

C. Device positions

Following the preliminary experiments outcome, devices were placed as illustrated in Fig. 4. The LoRa receiver was

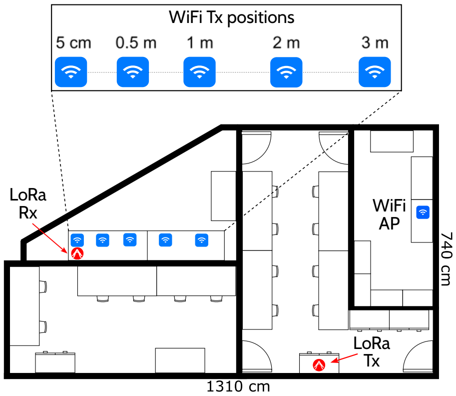


Fig. 4. Device positions during experiments. The LoRa devices and the WiFi Access Point (AP) were place at fixed positions, while the WiFi transmitter's position varied between 0 and 3m.

TABLE II
EXPERIMENT PARAMETERS.

Parameter	WiFi	LoRa
Radio channel	Channels 13, 14	2,479 MHz
Channel bandwidth	20 and 22MHz	812 KHz
Tx power	20 dBm	13dBm
Payload size	1540 Bytes	16 Bytes
Packet rate	1 pkt per ∼0.001s	1 pkt per ∼1s
WiFi mode	IEEE 802.11b/n	
Spreading Factors		5–12
Coding Rates		4/5–4/8

fixed at a reference position, while the WiFi transmitter (interference source) was placed at varying distances of 5 cm, 0.5 m, 1 m, 2 m, and 3 m. These distances were selected to simulate typical indoor use cases and to observe how interference decreases as the WiFi transmitter moves farther from the LoRa receiver. Preliminary tests revealed that beyond 3 m, additional distancing had minimal effect on packet reception, justifying the chosen upper bound.

The Access Point (AP) was positioned to minimize unintended interference and was configured to broadcast only beacon signals, as per IEEE 802.11 standards. The inclusion of both AP and a station device (STA) was necessary to establish a minimal operational WiFi link, allowing interference to be generated without transmitting excessive data. The placement of LoRa devices was guided by the need to maintain adequate link margin based on LoRa transmission characteristics.

D. Experiment Setup

The experimental environment consists of two independent wireless networks: (a) a LoRa communication link (one transmitter and one receiver), and (b) a WiFi network consisting of an AP and a STA. The WiFi network utilized IEEE 802.11 channel 13 (channel 14 for Direct-Sequence Spread Spectrum (DSSS) modulation scenarios). Key configuration parameters for both networks, including transmission power, packet sizes, modulation modes, and bandwidths, are detailed in Table II.

To quantify the communication robustness and link quality under interference-prone conditions, we measured the PRR over thousands of transmitted packets. Data from the Adafruit Feather V2 devices were collected via USB serial communication and logged by a Raspberry Pi Zero. Essential statistical measurements, including average PRR, SNR, and their respective variances were collected[2].

Environmental consistency was maintained by executing all experiments in a cleared space devoid of reflective surfaces or obstructions. A separate ESP32 device, operating in promiscuous mode, continuously monitored WiFi activity to confirm ongoing interference and automatically trigger re-association procedures in the event of disconnection. To improve experimental efficiency and mitigate device-induced latency, multiple pre-configured LoRa nodes were rotated across trials.

Each experiment lasted approximately 20 minutes—equivalent to 1000 LoRa transmissions—and followed a standardized sequence of steps to ensure consistency and repeatability. First, the devices were positioned according to the specific distance scenario, with antenna orientation kept fixed throughout. Next, the WiFi AP and STA were activated to generate interference. WiFi broadcast activity was then confirmed using a monitoring device. Once interference was verified, the LoRa receiver was started to receive packets. LoRa transmissions were subsequently initiated, and finally, the received data was recorded and all relevant logs were stored for analysis.

V. RESULTS AND DISCUSSION

In this section, we present the evaluation results and explore the implications of the LoRa-WiFi coexistence. The evaluation focuses on the PRR across all LoRa SFs (5–12), Coding Rates (CRs), and varying distances between the WiFi transmitter and the LoRa receiver. Two WiFi modulations were evaluated, OFDM and DSSS.

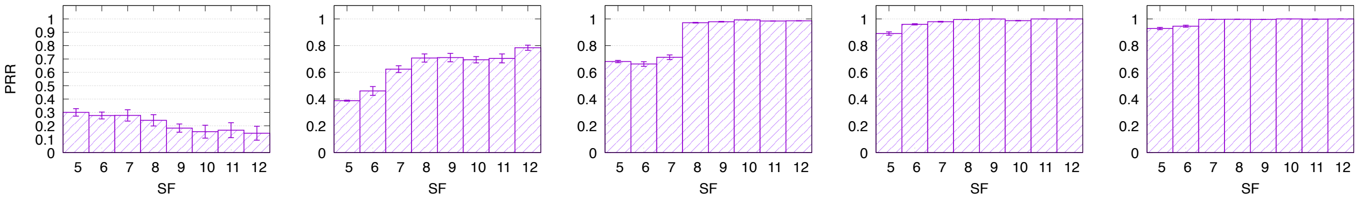
A. Packet Reception Ratio

Fig. 5 and 6 depict the PRR variations under OFDM and DSSS interference respectively across four spatial distances: 0 m, 0.5 m, 1 m, 2 m, and 3 m. When the WiFi transmitter is in close proximity to the LoRa receiver, particularly under OFDM interference, a significant degradation in PRR is observed at lower SFs. As SF increases, LoRa's CSS helps overcome the interference, gradually restoring PRR performance.

While DSSS-induced degradation exhibits a similar trend, its interference appears slightly less severe for equivalent SF and distance configurations. This discrepancy can be attributed to the narrower spectral footprint and lower peak power of DSSS compared to OFDM. The most obvious outcome is for the 5cm (i.e., 0 meters) scenario, where higher SFs exhibit much better performance in DSSS than in OFDM. For the rest of the cases, lower SFs ($\leq$9) are more negatively affected in DSSS than in OFDM.

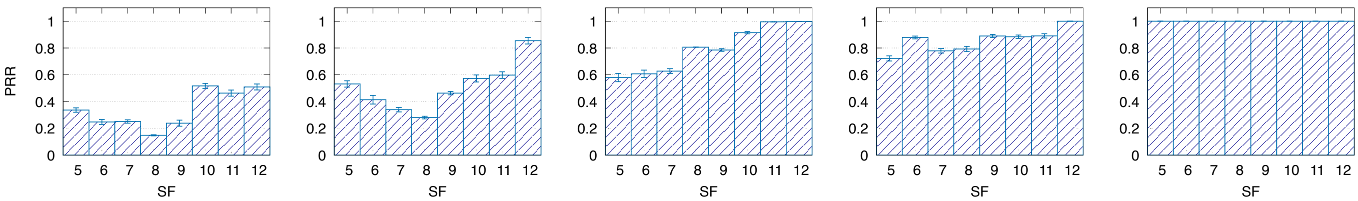
Fig. 7 shows the difference in RSS between the WiFi and LoRa signals measured at the LoRa receiver for all successfully received packets. By correlating these results with the PRR trends observed in the previous figures, we can

[2]All data will be archived at https://github.com/nu-iot-lab/lora2.4-interference-data



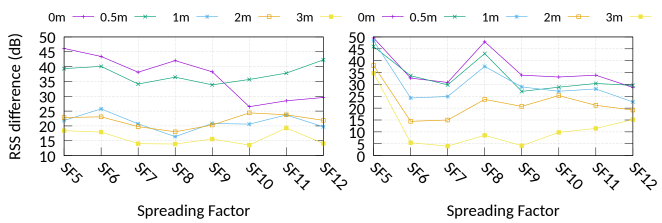
(a) 5 cm (b) 0.5 m (c) 1 m (d) 2 m (e) 3 m

Fig. 5. LoRa PRR for different SFs and distances to the WiFi transmitter with OFDM modulation.



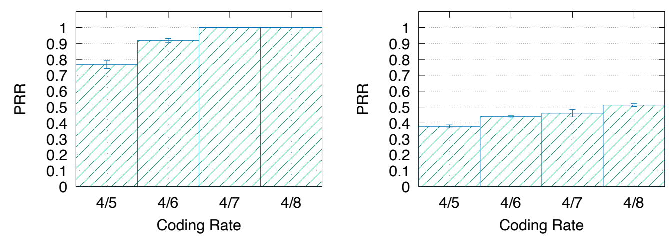
(a) 5 cm (b) 0.5 m (c) 1 m (d) 2 m (e) 3 m

Fig. 6. LoRa PRR for different SFs and distances to the WiFi transmitter with DSSS modulation.



(a) OFDM (b) DSSS

Fig. 7. Difference in received power between LoRa and WiFi signals for both simulations and all SFs.



(a) SF9, d=1m (b) SF5, d=0.5m

Fig. 8. PRR over variable CRs. Scenario (a): SF9 with 1 meter distance between the two devices; and (b): SF5 and 0.5m distance between them.

conclude that LoRa packets can still be successfully received even when the WiFi signal is 35–45 dB stronger. However, successful reception is more consistently observed when the RSS difference is around 25 dB, which appears to be a more reasonable average threshold. These results highlight LoRa's resilience to CTI in the 2.4 GHz band.

B. Coding Rate

In this subsection, we present results demonstrating the effect of using lower CRs (i.e., more error correction bits) to improve the PRR performance. Two indicative cases are employed; (a) SF9 with 1 meter distance between the WiFi transmitter and the LoRa receiver was employed; and (b) SF5 with a distance of half meter between devices. In both scenarios, we varied CR from the default value of 4/5 (weakest error correction) to 4/8 (strongest error correction).

Fig. 8 illustrates PRR for each available CR value for the two aforementioned cases. The results clearly indicate that a more robust CR yields a PRR increased by 23% and 12% in the SF9-1m and SF5-0.5m cases, respectively. This change suggests that increasing redundancy in the coding scheme helps to mitigate CTI.

VI. Conclusion & Future Work

The 2.4 GHz ISM band remains a popular choice for unlicensed IoT deployments due to its global availability and absence of duty cycle limits. However, coexistence with dominant technologies such as WiFi poses serious CTI challenges. This work experimentally evaluated the impact of such interference on LoRa performance in an indoor setting. Results showed that PRR improves significantly when the WiFi received power at the LoRa device is not higher than 25 dB, even though there are cases where packets can be successfully decoded even with 45 dB power difference. Low SFS (5–7) suffer most, while higher ones (10–12) maintain robustness. Increasing the coding rate from 4/5 to 4/8 also recovers up to 23% more packets. These findings emphasize that in the absence of coordinated medium access, LoRa performance is shaped largely by the interferer's proximity and activity. Building on this, future work will explore adaptive strategies, such as dynamic channel, SF, and coding rate adjustments.

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